Comparative analysis of the *Eulemur* vocal repertoire: a dynamic time warping approach.

Authors: Marco Gamba[a], Olivier Friard[a], Isidoro Riondato[a], Roberta Righini[a], Camilla Colombo[a], Longondraza Miaretsoa[b], Valeria Torti[a], Bakri Nadhurou[a], Cristina Giacoma[a].

[a] Department of Life Sciences and Systems Biology, University of Torino, Italy.
[b] Faculté de Sciences, University of Mahajanga, Madagascar.

Corresponding author: Marco Gamba, Department of Life Sciences and Systems Biology, University of Torino, Via Accademia Albertina 13, 10123 Torino, Italy.
Tel. +390116704560
Fax. +390116704508
Email: marco.gamba@unito.it

Appendix S1 Subjects, Study Sites, Equipment, Data Collection, Sampling, and Sound Analyses

Subjects, Study Sites, Equipment, and Data Collection

The recordings analyzed for the purpose of this study were part of a large collection of lemur sounds at the Department of Life Sciences and Systems Biology, University of Torino. The sound corpora originated from various recording campaigns focused on lemur vocal behavior in the wild and captivity. We collected spontaneous vocalizations occurring without visual displays or playback stimuli, using both *all-occurrence* and *focal animal* sampling methods (Altmann 1974). We also recorded vocalizations during veterinary inspections, which involved lemurs of all the study species, at the Parc Botanique et Zoologique de Tsimbazaza (Madagascar) in 2005. Recordings were carried out with Sony TCD-D100 and TCD-D8 digital tape recorders equipped with Sennheiser ME88 and Sony EMS-907 microphones. All microphones were equipped with slip-on foam windscreens. We used a sample rate of 48 kHz for all the recordings. We considered vocalizations emitted from adult or subadult individuals (> 2-year-old) only. All vocalizations emitted by infants and juveniles were discarded from the analysis.

Eulemur fulvus

The vocalizations of *E. fulvus* were collected from 28 individuals (17 males, 11 females). We recorded the vocal samples in the western dry forest of Ankatsabe-Analabe (16° 22′ 46.6″ S / 46° 29′ 02.0″ E; Mariarano, Mahajanga II District, Boeny region, Madagascar), in the humid forest of Maromizaha (18° 56′ 49″ S / 48° 27′ 53″ E; Anevoka, Alaotra-Mangoro region, Madagascar), and in the following zoos: Banham Zoo (United Kingdom), Colchester Zoo (United Kingdom), Linton Zoo (United Kingdom), Mulhouse Zoo (France), Parco Natura Viva – Guarding Zoological Park (Italy), and Parc Botanique et Zoologique de Tsimbazaza (Madagascar).

Eulemur rufus

The vocalizations of *E. rufus* were selected from three different corpora. The first corpus was derived from a recordings collected on 11 adult individuals (6 males, 5 females). The study was conducted from 2011 in five forest fragments of the Bombetoka-Belemboka forest (Mitsinjo district, 20 km southwest of Mahajanga, 15° 45′ 35″ S / 46° 30′ 48″ E): Antsakanala, Ankamaho, Amboaniokely, Analamavo, and Andohan'ny Sankoany. The second corpus comprises recordings from seven individuals (four males, three females). Vocalizations were recorded in Antsilahiza, a forest also located in the Mitsinjo district (80 km from Katsepy), in December, 2007. The third corpus was the result of recording sessions conducted in captivity on four individuals (two males, two females), at the Parc Botanique et Zoologique Tsimbazaza (Madagascar), in 2011.

Eulemur rubriventer

The audio recordings of *E. rubriventer* were selected from three different corpora. The first corpus comprised vocalizations of 16 individuals (7 males, 9 females), collected in 2008 in the Mulhouse Zoological and Botanical Garden (France). The second corpus comprised vocalizations from 24 individuals (15 males, 9 females) recorded in five European zoos in 1999: Apenheul Primate Park (Netherlands), Kölner Zoo (Germany), Linton Zoo (United Kingdom), Banham Zoo (United Kingdom), Twycross Zoo (United Kingdom). A third corpus comprised vocalizations recorded from seven captive individuals (five males, two females) at the Parc Botanique et Zoologique de Tsimbazaza (Madagascar) in 2005.

Eulemur mongoz

The dataset of *E. mongoz* calls comprises two corpora. The first corpus comprised vocalizations of 54 individuals (32 males, 22 females), collected from 2008 and 2102 in Madagascar and Comoros. We recorded the mongoose lemurs' vocalizations in Antsilahiza, (Mitsinjo District, 80 km from Katsepy; 16° 0′ 15.54″ S / 45° 52′ 18.81″ E, Madagascar), in the New Protected Area of Bombetoka-Belemboka (15° 45′ 35″ S / 46° 30′ 48″ E, Madagascar; Schwitzer *et al*. 2013), and in Anktsabe-Analabe forest (Mariarano 16° 22′ 46.6″ S / 46° 29′ 02.0″ E, Madagascar; Olivieri *et al*.

2005). All these sites are located in the Boeny region. In the Comoros, the vocalizations were recorded in Dziani (Anjouan Island, 12° 12′ 10.7″ S / 44° 28′ 44.2″ E), in Dzitso (Anjouan Island 12° 12′ 31.7″ S / 44° 28′ 09.8″ E), and in Ouallah (Mohéli Island, 12° 19' 23.9" S / 43° 39' 32.3" E). In addition, we also collected vocalizations from 19 captive individuals (twelve males, seven females) in 1999, in the following zoos: Mulhouse Zoo (France), Parc Botanique et Zoologique de Tsimbazaza (Madagascar), Banham Zoo (United Kingdom), Linton Zoo (United Kingdom), Colchester Zoo (United Kingdom), and Parco Natura Viva (Italy).

Eulemur coronatus

E. coronatus calls were recorded from 37 captive lemurs (27 males, 10 females) hosted in the following zoos: Mulhouse Zoo (France), Parc Botanique et Zoologique de Tsimbazaza (Madagascar), Banham Zoo (United Kingdom), Linton Zoo (United Kingdom), and Twycross Zoo (United Kingdom). Details can be found in Gamba and Giacoma (2007).

Eulemur flavifrons

The vocalizations of *E. flavifrons* were selected from a dataset of 37 individuals (22 males, 15 females) recorded in the following zoos: Apenheul Primate Park (Netherlands), Banham Zoo (United Kingdom), Linton Zoo (United Kingdom), Mulhouse Zoo (France), Parco Natura Viva – Guarding Zoological Park (Italy), and Parc Botanique et Zoologique de Tsimbazaza (Madagascar).

Eulemur macaco

The vocalizations of *E. macaco* were recorded from 21 individuals (13 males, eight females) recorded in the following zoos: Lemur Reserve (Madagascar), Parc Botanique et Zoologique de Tsimbazaza (Madagascar), Parco Natura Viva – Guarding Zoological Park (Italy), and St Louis Zoo (U.S.A).

Nomenclature

We have followed the nomenclature for each call type and the general classification that was defined by previous studies (Gamba and Giacoma 2007; Macedonia and Stanger 1994; Petter and Charles-Dominique 1979). The correspondence between nomenclature used in this article and those that were found in literature is provided in Table SI.

Sampling and Sound Analyses

Each vocalization was previously saved as a single file in WAV format (mono, 44100 Hz, 16 bit). Because vocalizations were taken from different corpora, the quality of the recordings was different due to background noise and distance of the lemurs to microphones (comprised between 3 and 15 meters). We visually screened all calls to ensure that we were selecting high-quality recordings. To limit the impact of background noise, we discarded low-quality calls (e.g. those where the signal to noise ratio was lower than 12 dB, which were acoustically distorted, and showed overlap with other sounds). Although all efforts have been made to orientate the microphone toward the mouth of the emitter, we cannot exclude that vocalization recorded with different orientation have entered the analyses.

This process led to datasets of different size (see ESM, Appendix S5). To ensure that we were passing on to the clustering analysis, all of the vocal types indicated in a previous quantitative investigation of the target species (Gamba 2006; Gamba and Giacoma 2005), MG, IR, RR, VT, and BN listened to all the vocalization and visually inspected the spectrograms to redefine the *a priori* classification as shown in Table I. We manually selected five calls for each identified vocal type. These five calls per putative vocal type were recorded from five different emitters (males or females). We then added a random sample of vocalizations of the same species up to 190 calls. This subsampling could effectively diminish the risk of having unreadable results from the clustering analysis and also limited the possibility that unrecognized vocal types would not have been

included in the analysis. We worked with datasets of equal size to avoid the possibility that different sizes may affect the estimation of vocal repertoire size.

Acoustic Dissimilarity Calculation

Pairwise sound comparison and efficient automatic or semi-automatic analytical procedures have seen an increasing use in the last years because of the diffusion of larger datasets (Clemins *et al.* 2006). Because time alignment of two sound patterns is a fundamental problem in speech recognition, the class of algorithms known as DTW has been developed to improve the accuracy of computer-based recognition systems (Davis and Mermelstein 1980). These systems ensure comparable results across different corpora. Sound dissimilarity indices were often indicated as a useful method for comparative approaches because they express the dissimilarity of two sounds using an objective and replicable form (Ranjard *et al.* 2010). Acoustic dissimilarity indexes have also been successfully used to study primate calls (e.g., Keenan *et al.* 2013; Lemasson *et al.* 2003, 2011).

In the present study, we used an implementation of DTW available in the software package DTWAVE (Ranjard *et al.* 2010), which includes the Hidden Markov Model Toolkit (Young 1994). We used a target rate of 15,000 ns, a window size of 30,000 ns, and a frequency range of 0 to 10,500 Hz. This frequency range included the salient features of most vocalizations, comprising best hearing sensitivity for most species (Ramsier *et al.* 2012), and reduced frequency bands typically affected by noise (e.g., insects, birds). Starting from the power cepstrum at a particular time the software processes the signal to build a vector. The result is called the mel-frequency cepstrum (Titze 1994), whose coefficients are called mel-frequency cepstral coefficients (MFCCs). When the cepstral coefficients were aligned and rescaled, and the average vector sequence was built, the software then calculated the pairwise distances between all the calls in the dataset (see Ranjard and Ross 2008). The result was a diagonally symmetric matrix for each species, in which each matrix element represented the pairwise dissimilarity between two of the calls.

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Appendix S2 Nomenclature Used for the Vocal Types

Table SI. The nomenclature used in this study for the vocal types of *E. fulvus, E. rufus, E. rubriventer, E. mongoz, E. coronatus, E. flavifrons, E. macaco* and the synonyms used in previous studies

Vocalization	***E. fulvus***	***E. rufus***	***E. rubriventer***	***E. mongoz***	***E. coronatus***	***E. flavifrons***	***E. macaco***
Alarm Call	Cuiiiiii ciot ciot[3].				Cuiiiiii ciot ciot[3]; Alarm Call [3,4].		Cuiiiiii ciot ciot[3]; Alarm Call[7].
Alarm Long Grunt				Long Grunt [1]; Creeeee[3] ; Ararar[3]; Long Creak[5].			
Chatter	Gurgle - Clear Call[1].			Contact-rejection call[3].			
Click	Clucking noise[3].			Click[1].	Click[4].		Clucking noise[3].
Group-Cohesion Call	Terrestrial disturbance/ advertisement call[1] ; crouiiiiiiiiiii[3].	Terrestrial disturbance/ advertisement call[1] .	Crouiiiiiiiiiii[3].			Terrestrial disturbance/ advertisement call[1].	Terrestrial disturbance/ advertisement call[1]; crouiiiiiiiiiii[3].
Grunted Hoot	Grunt-Hoot[1] Low-pitched grunt[3].	Grunt Hoot[1].			Grunt Hoot[1]; Grunted Hoot[4].	Grunted-Hoot[1].	Grunted-Hoot[1]; Grunted Hoot[7].
Grunt	Simple grunt[1]; Grunt[1]; Hard grunt[2] ; Purring[3] ; Hon[3].	Simple Grunt[1]; Grunt[1]	Short Grunt[1]; Grunt[1]	Short Grunt[1]; Closed/Open-Mouth Grunt [1]; Medium-intensity alarm call[3]; Hon[3]; Grunt agitated/soft[5].	Hard Grunt[1]; Grunt[4].	Grunt[1].	Grunt[1]; Short Grunt[2]; Purring[3]; Grunt[7].
Grunt-Tonal Call	Rising Grunt[1]; Contact-call[3].	Descending/ Rising Grunt[1].	Grunt Clear Call[1].		Rising Grunt[1]; Double Rising Grunt[1].		Modulated Long Grunt[2].
Gurgle			Gurgle[1].				
Hoot	Hoot[1]; Bark (?)[3].	Hoot[1].	Gulps/ Hoot[1].	Hoot[1].	Hoot [1,4].	Hoot[1].	Bark (?)[3]; Hoot[7].
Hoot-Tonal Call							
Long Grunt	Long Grunt[1].	Simple Grunt[1].	Long Grunt[1].	Long Grunt[1]; Greee[3]; Medium-intensity alarm call[3]; Creak[5].	Long Grunt [1,4].	Long Grunt[1].	Long Grunt[1]; Simple Long Grunt[2]; Long Grunt[7].

Vocalization	***E. fulvus***	***E. rufus***	***E. rubriventer***	***E. mongoz***	***E. coronatus***	***E. flavifrons***	***E. macaco***
Long Grunt-Tonal Call	Threat call: oe-ai-oe-ooo[3].			Grunt[1]; Threat call: oe-ai-oe-ooo[3].	Long Grunt Clear Call[4].		Threat call: oe-ai-oe-ooo[3]; Long Grunt Clear Call[7].
Scream	Terrestrial disturbance/ advertisement call[1] ; Kiou[3]; Distress Call[3].			Oum[3]; Distress Call[3]; Warble[5].	Scream[4].		Distress Call[3].
Snort-Grunt			Snort-Grunt[1].				
Snort-Grunt-Tonal Call			Snort-Grunt-Gurgle[1]; Snort-Grunt-Clear call[1]; Alarm Grunt[3]; Gre-ii[3].				
Snort	Sneeze[3].		Snort[1].	Snort[3]; Sneezing sound[3]; Snort[5].	Snort[4].		Sneeze[3]
Territorial Advertisement Call	Terrestrial disturbance/ advertisement call[1]; Creeeee[3]; Crou-crou-crou-crouiiiiiiiiiii[3].	Terrestrial disturbance/ advertisement call[1].	Terrestrial disturbance/ advertisement call[1].	Terrestrial disturbance/ advertisement[1]; Creeeee[3]; Screech[5].	Terrestrial disturbance/ advertisement call[1]; High intensity alarm call[3]	Terrestrial disturbance/ advertisement call[1].	Terrestrial disturbance/ advertisement call[1] ; Creeeee[3].
Tonal Call-Grunt							
Tonal Call	Group cohesion clear call[1]; Gurgle[1]; Meueu[3].		Group cohesion clear call[1].	Clear call[1].	Group cohesion clear call[1]; Gurgle[1]; Tonal Call[4].		Group cohesion clear call[1]; Tonal close contact call[1]; Trills[1]; Hon[3]; Clear Call[7].

(1) Macedonia and Stranger (1994);
(2) Grosset et al. (2001); (3) Petter and Charles-Dominique (1979); (4) Gamba and Giacoma (2007); (5) Curtis and Zaramody (1999); (6) Nadhouru et al. (2015);
(7) Pozzi et al. (2009). In Petter and Charles-Dominique (1979) some of the calls are referred to *Eulemur albifrons*.

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